

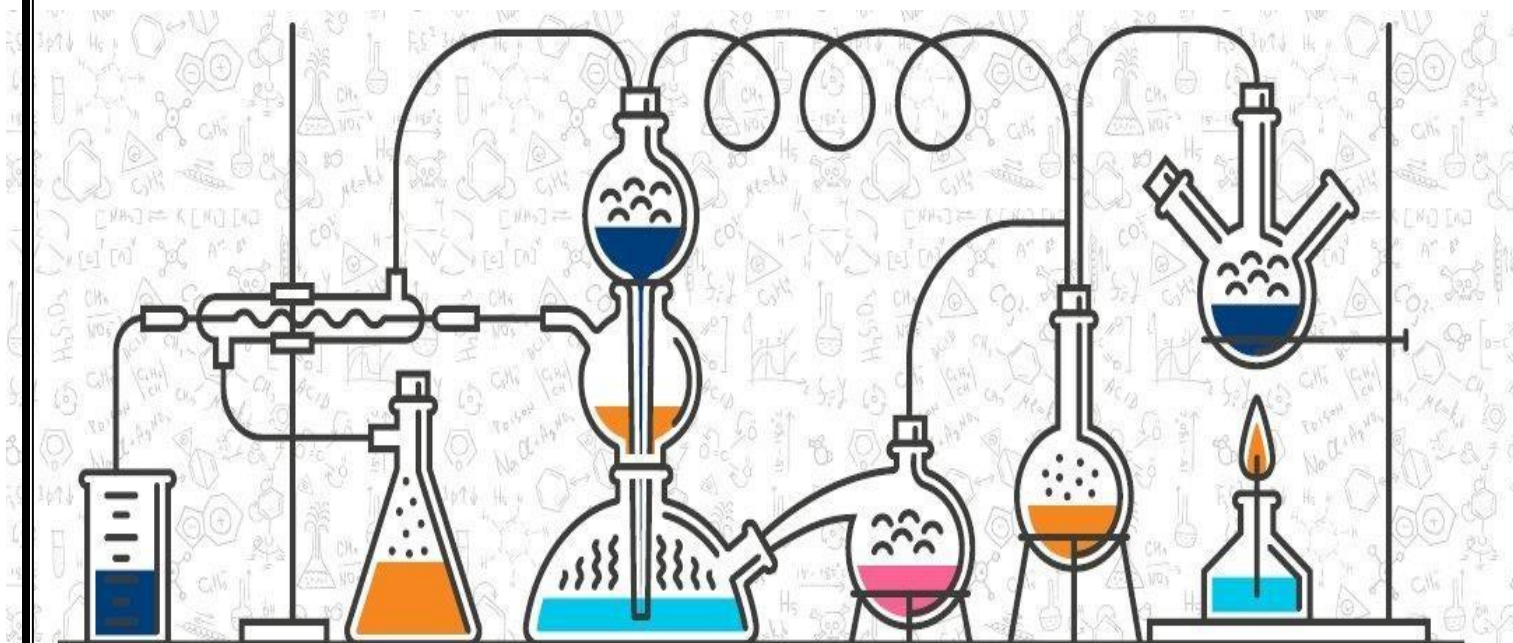


II SEMESTER (1BCHE102/202)

Smart Materials for Emerging Technology - CS Stream

Module – 2

Advanced Materials for Display Systems



DAYANANDA SAGAR COLLEGE OF ENGINEERING

**Accredited by National Assessment & Accreditation Council (NAAC) with 'A' Grade
(An Autonomous Institution affiliated to Visvesvaraya Technological University, Belagavi
& ISO 9001:2015 Certified)**

**DEPARTMENT OF CHEMISTRY
SHAVIGE MALLESWARA HILLS, KUMARASWAMY
LAYOUT BENGALURU-560111**

2025-26

Introduction to Nanomaterials

Nanomaterials describe materials of which a single unit is sized (in at least one dimension) between 1 and 1000 nanometers, but is usually 1-100 nm. Nanomaterials research takes a materials science-based approach to nanotechnology, advances in materials metrology and synthesis which have been developed in support of microfabrication research. Materials with structure at the nanoscale often have unique optical, electronic, or mechanical properties. A drastic change in the properties of nanomaterials can be observed when they are breakdown to the nanoscale level.

Size dependent properties of nanomaterials

Surface area:

- One of the most significant effects of reducing material size to the nanoscale is the drastic increase in surface-to-volume ratio.
- This high surface area increases the number of active sites available for chemical reactions or interactions with surrounding molecules.
- Therefore nanomaterial's can show specific surface related properties that are not observed in bulk materials.
- As a result, nanomaterials show enhanced adsorption, reactivity, and interaction with other species, making them highly effective in catalysis, sensors, drug delivery, and adsorption-based applications.
- For example, TiO₂ nanoparticles exhibit a much higher surface area than bulk TiO₂, which improves their photocatalytic efficiency. Bulk gold is catalytically inactive, but gold nanoparticles are catalytically very active for selective redox reaction.

Optical properties:

- The nanomaterial's of different size can scatter radiation of different wavelengths. Example: Color of few colloidal solutions are due to this scattering effect.
- Nano particles of metals exhibit unique optical property called as surface 'Plasmon resonance'.
- When lights hit the surface of metals particle, electron present on the surface (Surface plasmon) starts oscillating back and forth in a synchronized way in a small space and the effect is known as surface plasmon resonance.
- Depending on the frequency of oscillation resonating electrons capture radiation of different wavelength.
- For example, spherical gold nanomaterial of 25-nm diameter appears green while those in the range of 100 nm appear orange. Semiconductor quantum dots optical absorption and emission shift to the blue (higher energies) as the size of the dots decreases.

Conducting (Electrical) properties:

The electrical conductivity of nanomaterials depends on size, surface states, and electron transport mechanisms.

- The properties like conductivity come under category of electrical properties. The electronic bands in bulk material are continuous due to overlapping of orbital's of atoms. But in the nano size materials, very few atoms or molecules are present so the electronic bands become discrete. Hence, metals which are good conductors in bulk become semiconductors and/or insulators at na level.

- The reason is that the electronic bands in bulk material are continuous due to overlapping of orbitals of billions of atoms.
- Nanomaterial's very few atoms or molecules are present and so the electronic bonds become separate and separation between different electronic states varies with the size of nanomaterial

Classification of nanomaterials

The classification of nanomaterials is based on the number of dimensions. According to Siegel, nanostructured materials are classified as: zero dimensional (0D), one-dimensional (1D), two-dimensional (2D) and three dimensional (3D) nanomaterials.

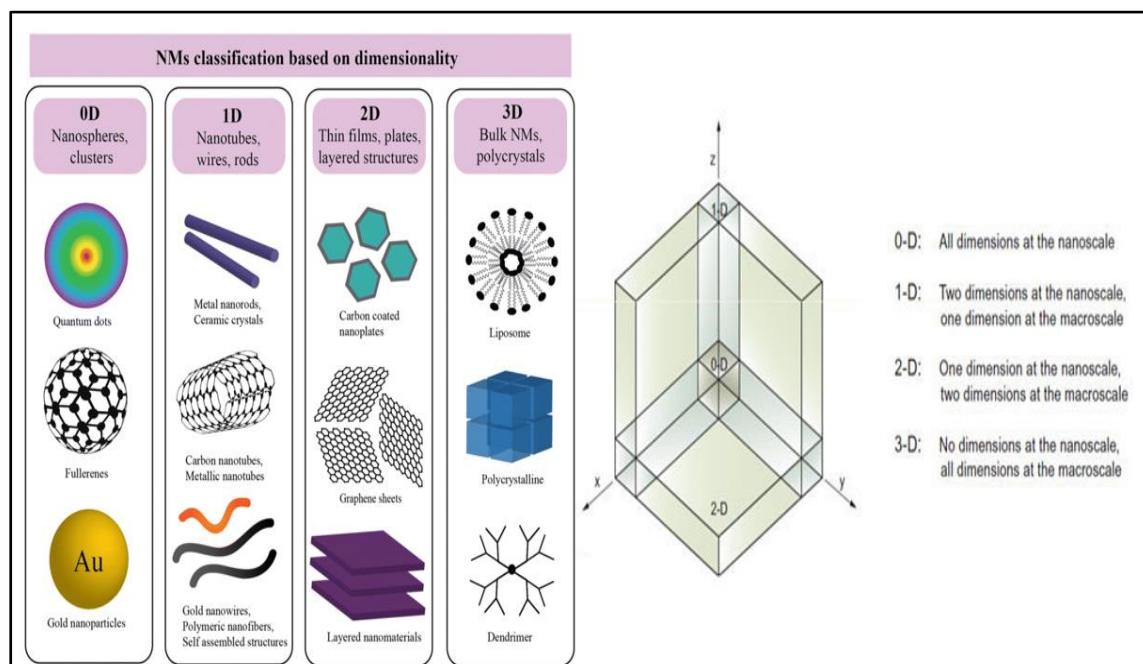


Fig 1: Nanomaterial classification

- ❖ Zero-dimensional nanomaterials (0D): Here, all the dimensions (x, y, z) are measured within the nanoscale (no dimensions are larger than 100 nm). Most commonly, 0D nanomaterials are nanoparticles such as quantum dots and nanospheres.
- ❖ One-dimensional nanomaterials (1D): Here, two dimensions (x, y) are at the nanoscale, and the other is outside the nanoscale. This leads to needle-shaped nanomaterials. It includes nanofibers, nanotubes, nanorods, and nanowires.
- ❖ Two-dimensional nanomaterials (2D): Here, one dimension (x) is at the nanoscale and the other two are outside the nanoscale. The 2D nanomaterials exhibit platelike shapes. It includes nanofilms, nanolayers and nanocoatings with nanometer thickness.
- ❖ Three-dimensional nanomaterials (3D): These are the nanomaterials that are not confined to the nanoscale in any dimension. These materials have three arbitrary dimensions above 100 nm. The bulk (3D) nanomaterials are composed of a multiple arrangement of nanosized crystals in different orientations. It includes dispersions of nanoparticles, bundles of nanowires and nanotubes, as well as multi-nanolayers (polycrystals).

Classification of nanomaterials:

Based on Composition: Nanomaterials can be categorized into four types.

- ❖ **Inorganic-based nanomaterials:** An inorganic nanomaterial (NM) can comprise a metal or non-metal element, or take the form of an oxide, hydroxide, chalcogenide, or phosphate compound. Example: Examples of metal-based and metal oxide based inorganic nanomaterials are silver (Ag), gold (Au), zinc oxide (ZnO), and copper oxide (CuO).
- ❖ **Carbon-based nanomaterials:** Carbon-based are amazing technological advancement with unique properties (high mechanical strength, high conductivity, attractive optical properties, chemical versatility, etc.). Carbon-based nanomaterials include graphene, fullerene, single-walled carbon nanotube, multiwalled carbon nanotube, carbon fiber, an activated carbon, and carbon black.
- ❖ **Organic-based nanomaterials:** The organic-based nanomaterials are formed from organic materials excluding carbon materials. For example, dendrimers, cyclodextrin, liposome, and micelle.
- ❖ **Composite-based nanomaterials:** Composite nanoparticles are defined as the nanomaterials with composite structure which are constituted by two or more components of nanoscale with special physical and chemical properties. The composite nanomaterials are any combination of metal-based, metal oxide-based, carbon-based, and/or organic-based nanomaterials, and these nanomaterials have complicated structures like a metal-organic framework. For example, ZnO-Cu₂O, Al-SiC composites.

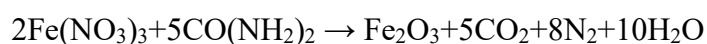
Synthesis of Fe₂O₃ nanoparticles by solution combustion method:

The solution combustion method is a rapid and energy-efficient technique for synthesizing metal oxide nanoparticles. In this method, a metal nitrate (oxidizer) reacts with an organic fuel (such as urea, glycine, or citric acid) in an aqueous solution, leading to a self-sustained exothermic redox reaction that produces fine oxide powders.

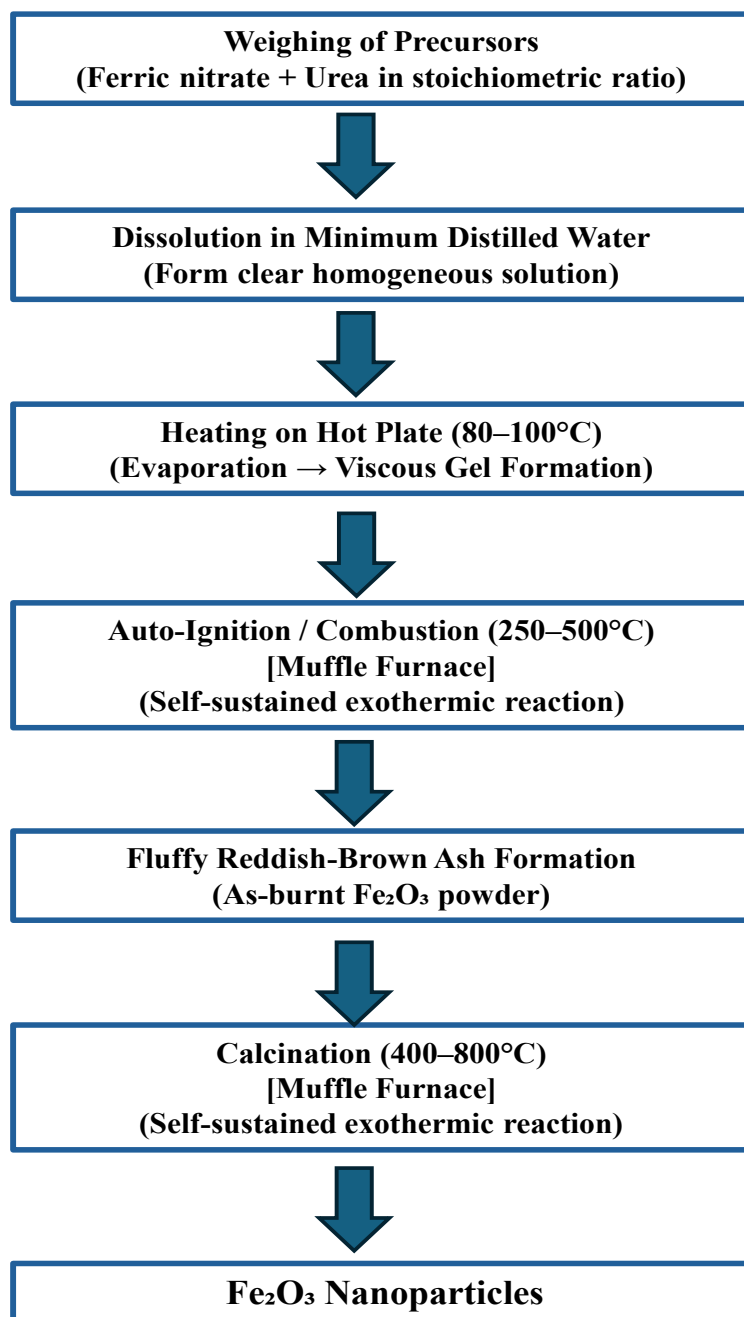
Chemicals used

Fe(NO₃)₃ → Oxidizer (provides oxygen) Urea → Fuel (reducing agent)

Combustion Reaction



Flow Chart



Quantum Dots

Nanometer-sized semiconductor particles called quantum dots, which are usually between 2 and 10 nm in size, confine electrons in all three spatial dimensions. This creates quantum confinement phenomena, which give the particles special optical and electrical characteristics like size-dependent light emission. Since their behavior, they are frequently called "artificial atoms" since they have distinct energy levels that successfully bridge the gap between single molecules and bulk semiconductors. The colour (wavelength) of light emitted by a QD depends on its size: smaller QDs emit blue light (shorter wavelength), larger QDs emit red (longer wavelength).

Properties:

- Size-Dependent Emission
- Quantum Confinement
- Tunable Band Gap

Applications:

- Display and lighting
- Solar Cells
- Environmental applications
- Quantum Computing
- Sensing and Biological imaging

Inorganic Quantum Dot materials- (IQDMs)

These types of QDMs are made from inorganic semiconductors, such as CdSe, PbS, InP, and ZnO. IQDMs are very tiny crystals of inorganic semiconductors (2–10 nm in size) whose optical and electronic properties can be tuned by changing their size, due to the quantum confinement effect.

Synthesis and properties of silicon based QDs by sol gel method

1.Preparation of sol:

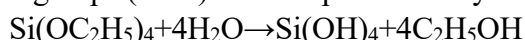
The sol–gel process begins with silicon-containing precursors known as silicon alkoxides. The most common precursors are:

- Tetraethyl orthosilicate (TEOS) – $\text{Si}(\text{OC}_2\text{H}_5)_4$
- Tetramethyl orthosilicate (TMOS) – $\text{Si}(\text{OCH}_3)_4$

These molecules contain Si–OR groups (R = alkyl), which are reactive toward water. The precursor is typically dissolved in an alcohol medium and mixed with a catalyst (acid or base) to control the reaction rate.

2.Conversion of sol to gel:

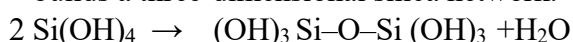
In the presence of water, the alkoxide groups (–OR) on the silicon precursor are replaced by hydroxyl groups (–OH). This step is called hydrolysis.



Hydrolysis produces silanol groups (Si–OH), which are essential for the next step. The rate of hydrolysis depends on pH, temperature, and water-to-alkoxide ratio.

3.Condensation Reactions

After hydrolysis, the silanol groups condense to form Si–O–Si linkages. This polymerization process builds a three-dimensional silica network.



Step-4: Drying

The gel is dried to obtain a solid silica network.

Step-5: Annealing

At high temperature (e.g., 800–1100 °C) in a reducing atmosphere (H_2), Si-QDs nucleate and grow inside the silica matrix. Further, Si can be purified from the matrix if needed.

Properties & Applications of inorganic quantum dots

- **Tunable optical properties:** By precisely controlling the size of the quantum dot during

synthesis, the color of the emitted light can be tuned from the ultraviolet to the infrared spectrum.

- **High brightness and photostability:** highly bright and resistant to photobleaching,
- **High quantum yield:** They have a high efficiency in converting absorbed light into emitted light, often much higher than traditional organic dyes.
- **Surface properties:** Their surface can be modified with ligands or polymers to improve water solubility, reduce toxicity, and enable bio-conjugation for specific targeting.

Applications

- **Displays and lighting:** Quantum dot displays (QD-LEDs) offer superior color accuracy, efficiency, and brightness compared to traditional displays. They are also used in solid-state lighting to improve colour quality.
- **Solar cells:** QDs are used in both "quantum dot only" and hybrid solar cells to improve efficiency in converting sunlight into electricity.
- **Biomedical imaging:** They can be used as fluorescent labels for high-sensitivity bio-imaging, such as in confocal microscopy and for tracking cells. Their stability and bright emission are ideal for this purpose.
- **Drug delivery:** Functionalized quantum dots can be used to deliver drugs to specific cells and tissues, with their fluorescence used to monitor the delivery process.
- **Sensing:** QDs can be incorporated into biosensors and electrochemical sensors for detecting a wide range of analytes, including heavy metals and biomolecules.
- Quantum dots are also being explored for use in photocatalysis (e.g., water splitting), quantum computing, and advanced computing technologies.

ORGANIC QUANTUM DOT MATERIALS (OQDMS)

Organic Quantum Dot materials-(OQDMs):

Quantum Dots that are created from organic materials (such carbon, graphene, conducting polymers, or organic compounds). Examples include carbon quantum dots (CQDs), graphene quantum dots (GQDs), and polymer-based QDs

Synthesis of Graphene Quantum Dots using citric acid method:

Graphene Quantum Dots (GQDs) are tiny pieces of graphene that are less than 20 nm in size and exhibit unique properties in specific conditions. These can be synthesised through the citric acid method, which is a simple and cost-effective "bottom-up" approach using a common citric acid organic compound.

Step-1: Take a definite quantity of citric acid (around 1–2 grams) and transfer it to a dry, clean beaker or crucible.

Step-2: Heating/Pyrolysis

Melt the citric acid on a hot plate or in an oven at 200–250°C for around 10–30 minutes. The citric acid initially melts, and then slowly becomes orange or brown, showing the production of small carbon-based particles (GQDs).



Step-3: Cooling and Dissolving

Let the hot melt cool down to room temperature. Dissolve the solidified material by adding distilled water, creating a GQD solution.

Step-4: Purification

Dispose or dialyse the solution to eliminate any residual unreacted citric acid or larger impurities. The resulting solution should emit blue glow under UV light due to the fluorescent nature of GQDs.

Properties:

- ❖ Non-Toxic & Biocompatible: Carbon/organic QDs are safer compared to heavy-metal inorganic QDs.
- ❖ Photoluminescence (Fluorescence): Strong light emission, tunable by size and surface groups.
- ❖ Solution Processability: Easily dispersed in water and organic solvents → good for flexible/printable devices.
- ❖ Functionalization: Surface can be chemically modified to improve performance.
- ❖ Eco-friendly & Low-cost: Derived from organic/biological materials.

Applications:

- ❖ LEDs
- ❖ Photovoltaics
- ❖ Used in next-generation flash memory devices
- ❖ GQDs are incorporated into field-effect transistor (FET) due to their high electron mobility and adjustable bandgap.

Used in conductive inks for printed electronics

Display system

Display system: A Display system is a system through which information is conveyed to people through visual means. Display System means a system for the use of Display Materials. The display systems consist of Photoactive and electroactive materials as the active materials, which convert the electrical energy into light emissions. These materials have unique properties, such as piezoelectricity or electroluminescence, that allow them to be used in a variety of display technologies. They provide unique features such as high contrast and low power consumption.

Photoactive material: materials that change their properties when exposed to light, often in the ultraviolet or visible region of the electromagnetic spectrum. Example: conducting polymers such as polythiophene, polypyrrole and nanomaterials such as TiO₂, ZnS etc.

Electroactive materials: An electroactive material exhibits a change in size or shape when stimulated by an electric field. Example: polyacetylene, Barium Titanate, Potassium Niobate etc.

Applications of Photo and Electroactive materials in optoelectronic devices:

Photoactive and electroactive materials belong to the huge field of photonics, where materials that actively interact with light are tuned and optimized to achieve effects such as;

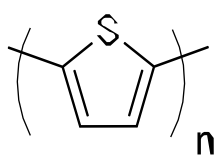
- Convert electrical signal to light- light emission (LEDs and lasers, just to name the most common ones)
- Light detection, with related signal amplification (e.g., in photomultipliers) and processing operations.
- Alternatively, they can be used to develop light-sensitive circuits and switches (such as with photoresistors),
- Convert light into an electrical signal (photodiodes – PV Cells)

Nano-materials and organic materials used in optoelectronic devices

1. Organic materials as optoelectronic devices:

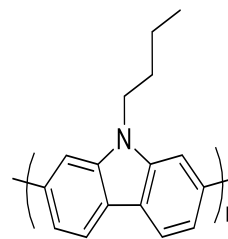
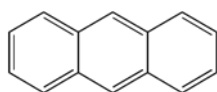
Organic molecules used as optoelectronic devices: small molecules and conjugated polymers, the two main types of organic materials used for optoelectronic and photonic devices, can be used in several applications, including organic light-emitting diodes, photovoltaic devices, etc. Organic molecules having alternative conjugation with planar sp² carbon in the backbone structure are basically used for optoelectronic applications. These conjugated materials, hence, behave as semiconductor materials. Anthracene is one such molecule known for almost a century (1910) possesses alternative conjugation and a planar structure.

Organic semiconductor materials in general can be separated into two classes: conjugated polymers and small molecules.



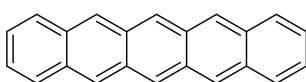
Polythiophene

Anthracene

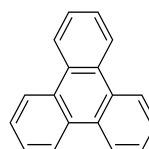


Polycarbazole

Polymers have high molecular weights that result good mechanical and thermal stability, and hence, polymers are advantageous for large-area device fabrications. Example: polythiophene, polycarbazole. Small molecules have advantages in good batch-to-batch repeatability, high crystallinity, and excellent charge carrier mobilities due to strong intermolecular π - π stacking. Example: pentacene, triphenylene



pentacene



triphenylene

2. Nano materials as optoelectronic devices

- ❖ Optical properties such as absorption, transmission, reflection, and light emission are exhibited by nanomaterials.

- ❖ Intrinsic characteristics of nanoparticles such as size, shape, and surface properties determine its applications.

- ❖ At a specific wavelength (frequency) of light, the collective oscillation of electrons on the metal nanoparticle surface causes a phenomenon called surface plasmon resonance, which results in a strong extinction of light which making the nanomaterial a materials for optical and optoelectronics applications.

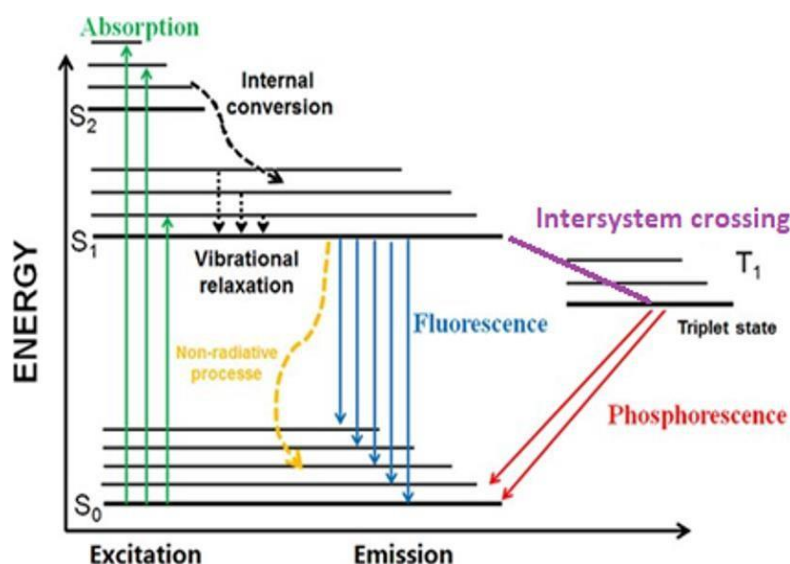
- ❖ Metal oxides such as TiO₂ and ZnO and quantum dots such as CdSeS and ZnS are used for optoelectronic applications.

Jablonski diagram

The Jablonski diagram is widely used in fluorescence spectroscopy to illustrate the excited states of a molecule and the radiative and non-radiative transitions that can occur between them. Light absorption is the process in which light is absorbed by matter and converted into energy. Once a molecule has absorbed energy in the form of electromagnetic radiation, there are several routes by which it can return to ground state.

The naming of the electronic states is based on the spin angular momentum configuration of each state. Singlet states (a total spin angular momentum of zero) are denoted by an S and triplet states (a total spin angular momentum of one) by T:

- S_0 is the singlet ground state of the molecule
- S_1 is the first excited singlet state
- T_1 is the first excited triplet state



The two main routes are: radiative process and nonradiative process.

- ❖ **Radiative process:** It involves the emission of the absorbed energy as radiation or photon.
 - **Fluorescence:** If the photon emission occurs between states of the same spin state (e.g. $S_1 \rightarrow S_0$) this is termed fluorescence. Since the transition is between similar state, it is allowed transition and lifetimes of fluorescent states are very short (1×10^{-5} to 10^{-8} seconds).
 - **Phosphorescence:** If the spin state of the initial and final energy levels is different (e.g. $T_1 \rightarrow S_0$), the emission (loss of energy) is called phosphorescence. This a forbidden transition and hence, fluorescence is statistically much more likely than phosphorescence for most molecules. Phosphorescence somewhat longer (1×10^{-4} seconds to minutes or even hours; think about glow-in-the-dark flying disks).
- ❖ **Nonradiative process:** Three nonradiative deactivation processes,
 - **Internal conversion (IC):** Internal conversion is the radiation less transition between energy states of the same spin state (compare with fluorescence-a radiative process).
 - **Intersystem crossing (ISC):** Intersystem crossing is a radiation less transition between different spin states (compare to phosphorescence).
 - **Vibrational relaxation:** Vibrational relaxation, the most common of the three--for most molecules,

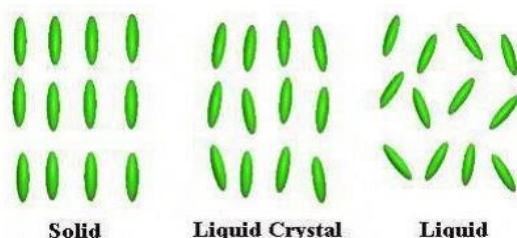
occurs very quickly ($<1 \times 10^{-12}$ seconds) and is enhanced by physical contact of an excited molecule with other particles with which energy, in the form of vibrations and rotations, can be transferred through collisions. This means that most excited state molecules never emit any energy because in liquid samples the solvent or, in gas phase samples, other gas phase molecules that are present "steal" the energy before other deactivation processes can occur.

Liquid crystals

Introduction

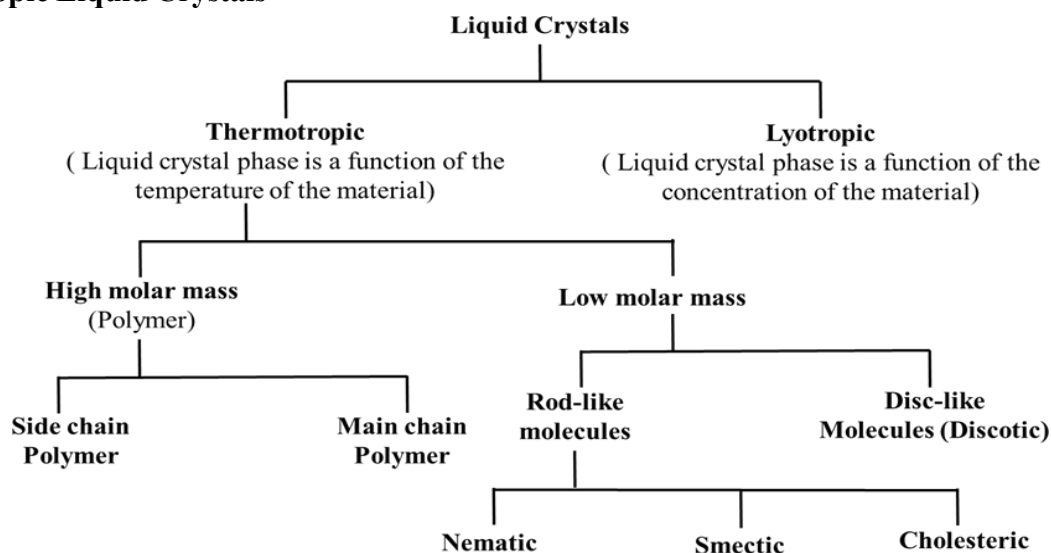
LCs are a unique state of matter, between solid (crystalline) and liquid (isotropic) phases some compounds form a distinct, different intermediate phase, sometimes referred to as the "fourth state of matter" or "mesophase." In crystalline solids, molecules exhibit both positional and orientational order and hence, they are anisotropic in nature. In liquids, molecules exhibit neither positional nor orientational order and hence, isotropic in nature. In a liquid crystal, the molecules can move around and flow like a liquid, but they maintain a certain degree of alignment, like a solid crystal. Hence, the phase is called "mesophase" and molecules are referred to as mesogens.

- ❖ Also possess properties characteristics not found in either solids or liquids.
- ❖ may response to external perturbation and some changes colour with temperature



Classification of Liquid crystals

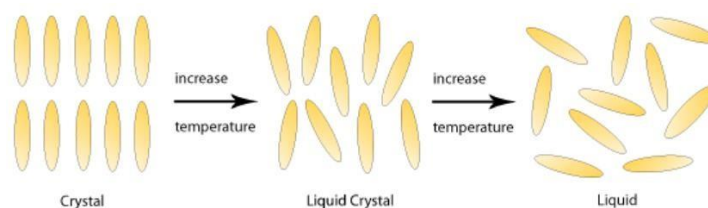
Lyotropic Liquid Crystals



Some of the compounds transformed into liquid crystal phase when mixed with another substance or solvent by the variation of concentration of the compound are called lyotropic liquid crystals. Ex: Soap water mixture

Thermotropic liquid crystals

The materials that exhibit liquid crystalline behavior on variation of temperature alone are referred to as thermotropic liquid crystals. Their liquid crystalline phase occurs within a limited temperature range between the solid and liquid phase.



Examples of thermotropic liquid crystals:

Cholesteryl Benzoate [Liquid crystal range: 145.5°C –178.5°C]

Diethyl p-terphenyl-p-p''-carboxylate [Liquid crystal range: 173⁰C –189⁰C]

Thermotropic liquid crystals are further classified as: There are three types of liquid crystals

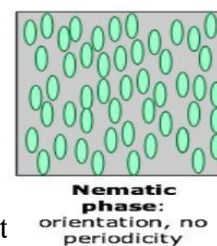
1.Nematic Liquid Crystals (NLC)

2.Smectic Liquid Crystals (SLC)

3.Chiral Liquid Crystals or Cholesteric Liquid Crystals (CLC)

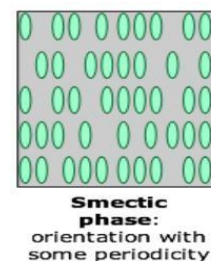
Nematic Liquid Crystals (NLC)

- ❖ These are formed by the compounds that are optically inactive.
- ❖ The nematic phase has calamitic or rod-shaped organic molecules that have no positional order.
- ❖ The molecules have elongated shape and are oriented parallel to the director.
- ❖ These molecules possess intermolecular force of the attraction such that they stay parallel to one another to form nematic liquid crystals.



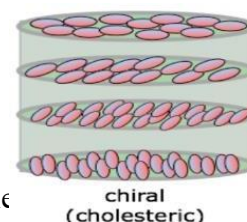
Smectic Liquid Crystals:

- ❖ Smectic liquid crystals are a type of liquid crystals that have well-defined layers of molecules that can slide over one another. This behaviour is like the sliding effects given by soap.
- ❖ These liquid crystals have small amount of positional order and orientational order.
- ❖ smectic liquid crystals occur at temperatures lower than that of the nematic liquid crystals.



Cholesteric Liquid Crystals:

- ❖ Cholesteric liquid crystals are a type of liquid crystals made only of chiral molecules. This type of phase shows chirality.
- ❖ The term cholesteric comes from its first observation where this phase of matter first appeared in cholesterol derivatives.
- ❖ We can observe this type of phase exhibiting a twisting of the molecules perpendicular to the director where the molecular axis is parallel to the director.



Properties and applications of Liquid Crystal Display (LCD):

Properties:

- 1.**Electro-Optical Effect:** LCDs modify the optical properties of liquid crystal molecules to realign them with a tiny electric field, modulating light passage through each pixel.
- 2.**Low Power Consumption:** Because LCDs only require low voltages to operate, they consume a lot less energy than more outdated technologies like CRTs.
- 3.**Lightweight and Thin:** LCDs are ideal for portable and wall-mounted devices because to their structure, which makes them extremely light and thin.
- 4.**No Direct Light Emission:** Liquid crystals don't emit light on their own. They need backlighting from LED or fluorescent lights or external reflection.
- 5.**Outstanding Display Quality:** Because of its excellent contrast, sharpness, and colour reproduction, LCD panels offer high-quality images for a variety of devices.

Applications:

- 1.**Display Screens:** The most common use is in screens for gadgets like digital watches, laptops, TVs, smartphones, and calculators. LCDs are also utilised in instrumentation panels and medical devices.
- 2.**Medical Imaging:** Liquid crystals are used in specialised instruments, including thermography systems, to identify temperature changes in the human body in order to make diagnoses.
- 3.**Optical Devices:** By precisely manipulating light, LCD technology is used in lenses, filters, and projection systems.
- 4.**Consumer Electronics:** Because of their robustness and energy efficiency, these devices are frequently found in mobile phones, portable video games, and automobile navigation systems.
- 5.**Industrial and Scientific Instruments:** Used in spectrum analysers, oscilloscopes, and other control panels that need a good visual readout.

Properties and applications of Light Emitting Diodes (LED):

Properties:

1. **Semiconductor Device:** LEDs are forward biased p-n junction semiconductor diodes that produce light.
2. **Electroluminescence:** LEDs transform electrical energy into light energy by means of electron-hole recombination, which is the basis of their operation.
3. **Energy Efficiency:** Compared to traditional lights, they are incredibly energy efficient, using less power and generating less heat.
4. **Quick Response:** When a forward current is applied, LEDs can turn on very quickly and light up immediately.
5. **Extended Lifespan:** Compared to fluorescent or incandescent lights, LEDs have a far longer operating lifespan.
6. **Monochromatic Light:** The wavelength (colour) of light emitted by each LED is dictated by the semiconductor material and its band gap.

Applications:

- 1.**Indication Lamps:** These are frequently used as indication lights on panels and electronic gadgets.
- 2.**Display Screens:** Found in seven-segment displays, LED TVs, digital displays, and calculators.
- 3.**Illumination:** Because of their great longevity and effectiveness, they are frequently utilised for street, business, and residential lighting.
- 4.**Automotive Lighting:** Used in dashboard indications, brake lights, and headlights for vehicles because of its durability and fast response.
- 5.**Traffic Signals:** Because they are bright and reliable, traffic signals use LED.

Properties and applications of Organic Light Emitting Diodes (OLED):

Properties:

Some of the key properties of Organic Light Emitting Diodes (OLEDs) include:

1. **Thinness and flexibility:** OLEDs are very thin and flexible, which makes them suitable for use in curved or flexible displays.
2. **High contrast:** OLEDs have a high contrast ratio, which means that they can produce deep blacks and bright whites, resulting in images with vivid and rich colours.
3. **Fast response time:** OLEDs have a fast response time, which means that they can switch on and off quickly, resulting in smooth and seamless motion in video content.
4. **Wide viewing angle:** OLEDs have a wide viewing angle, which means that the image quality is maintained even when viewed from different angles.
5. **Energy efficiency:** OLEDs are energy efficient as they do not require a back light like traditional LCD displays, resulting in lower power consumption.

Applications

- ❖ Flat-panel TV screen
- ❖ Digital cameras
- ❖ Mobile phones

Properties and applications of Quantum Light Emitting Diodes (QLED):

Properties:

QLED is an electroluminescent device that uses quantum dots (QD's) as a source of light Emission.

1. **Accurate and vibrant colours:** QLEDs can produce highly accurate and vibrant colours due to their use of quantum dots, which emit light of a specific colour when they are excited by a light source or an electrical current.
2. **Energy-efficient:** QLEDs are more energy-efficient than traditional LCD displays because they do not require as much back lighting.
3. **High contrast:** QLED displays have high contrast ratios, which means that the difference between the darkest and brightest areas of the display is greater, resulting in more detailed and life like images.
4. **Long life span:** QLEDs have a longer life span than traditional LCD displays because they do not suffer from the same issue soft back light burnout or colour fading over time.
5. **Fast response times:** QLED displays have fast response times, which mean that they can display fast-moving images without motion blur or ghosting.
6. **Flexibility:** QLEDs can be made on flexible substrates, which allows for the creation of flexible displays that can be bent or curved.

Applications

- ❖ Flat-panel TV screen
- ❖ Digital cameras
- ❖ Mobile phones

Question Bank:

4 marks

1. Write the classification of nanomaterials based on dimension with examples.
2. Write the classification of nanomaterials based on composition with examples
3. Explain the size-dependent properties of nanomaterials.
4. List any properties and applications of quantum dots.
5. Write the properties and applications of inorganic quantum dots with examples
6. What are organic quantum dots (OQDMs)? Give an example.
7. Write the properties and applications of organic quantum dots with examples
8. Define graphene quantum dots (GQDs). Write their properties.
9. List any two properties and applications of graphene quantum dots.
10. What are photoactive materials? Give two examples.
11. Discuss the materials used in optoelectronics with examples.
12. What are electroactive materials? Give examples.
13. Write the properties and applications of Liquid Crystal Display (LCD).
14. Write the properties and applications of Liquid Emitting Display (LED).
15. Write the properties and applications of Organic Light Emitting Diodes (OLED's).
16. Write the properties and applications of Quantum Light Emitting Diodes (QLED's)

6 and 8 marks

1. Explain the size-dependent properties of nanomaterials.
2. Describe the solution combustion method for the synthesis of Fe_2O_3 nanoparticles with reactions and flow diagram.
3. Explain the sol-gel method for the synthesis of silicon-based quantum dots and discuss their properties.
4. Describe the citric acid method for the synthesis of graphene quantum dots.
5. Explain the Jablonski diagram and discuss various transitions involved in it.
6. Discuss the types of liquid crystals and their characteristics.
7. Define inorganic quantum dots. Write their properties and applications.
8. Define organic quantum dots. Write their properties and applications.